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(19) **United States**(12) **Patent Application Publication**
UEDA et al.(10) **Pub. No.: US 2020/0131739 A1**(43) **Pub. Date: Apr. 30, 2020**(54) **WHEEL LOADER AND BUCKET LOAD
CALCULATION METHOD****Publication Classification**(71) Applicant: **HITACHI CONSTRUCTION
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AOKI**, Tsukuba (JP)(51) **Int. Cl.**
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G01G 19/16 (2006.01)(52) **U.S. Cl.**
CPC **E02F 9/26** (2013.01); **E02F 3/283**
(2013.01); **G01G 19/16** (2013.01); **G01G
19/10** (2013.01); **G07C 5/02** (2013.01)(57) **ABSTRACT**

In an unloaded state of the bucket (3), a control device (30) calculates an unloaded moment (M_0) around a hinge pin G from the pressure of the lift arm cylinder that is detected by pressure sensors (15a, 15b), the angle of a lift arm that is detected by a lift arm angle sensor 14, and dimension data extracted from a vehicle body information database (35). In a loaded state of the bucket, the control device calculates a loaded moment (M_1) around the hinge pin G from the pressure of the lift arm cylinder, the angle of the lift arm, and the dimension data extracted from the vehicle body information database. The control device calculates the load (W) on the bucket by dividing the difference between the unloaded moment and the loaded moment by the horizontal distance (L_w) between the hinge pin and the gravity center position of the bucket.

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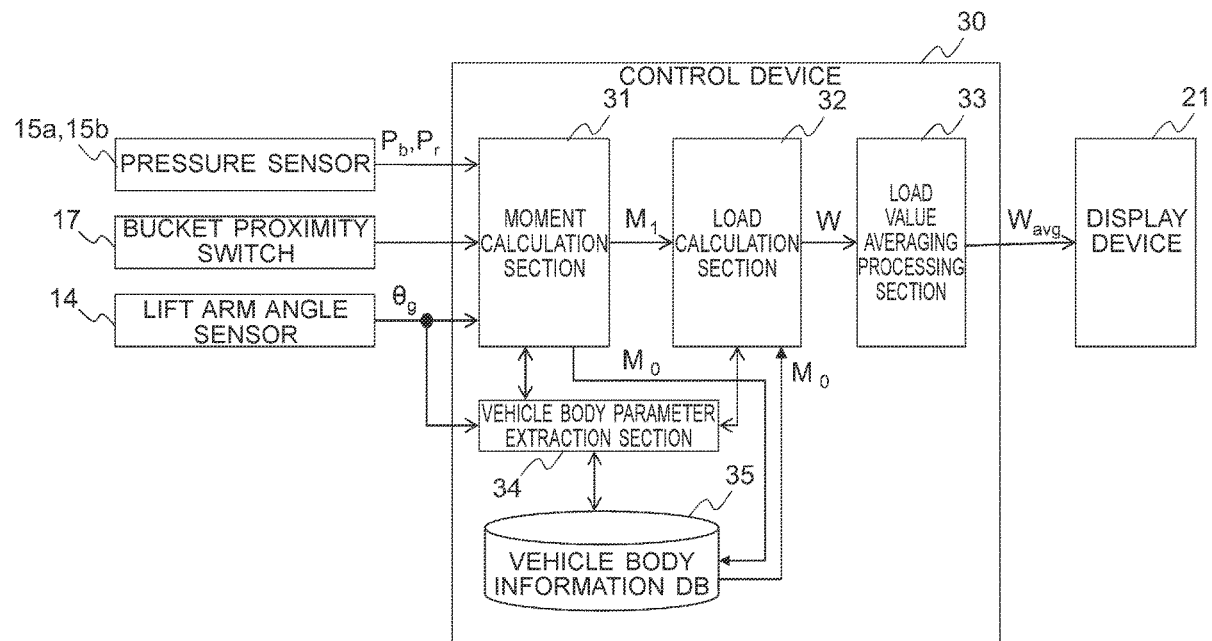


FIG. 1

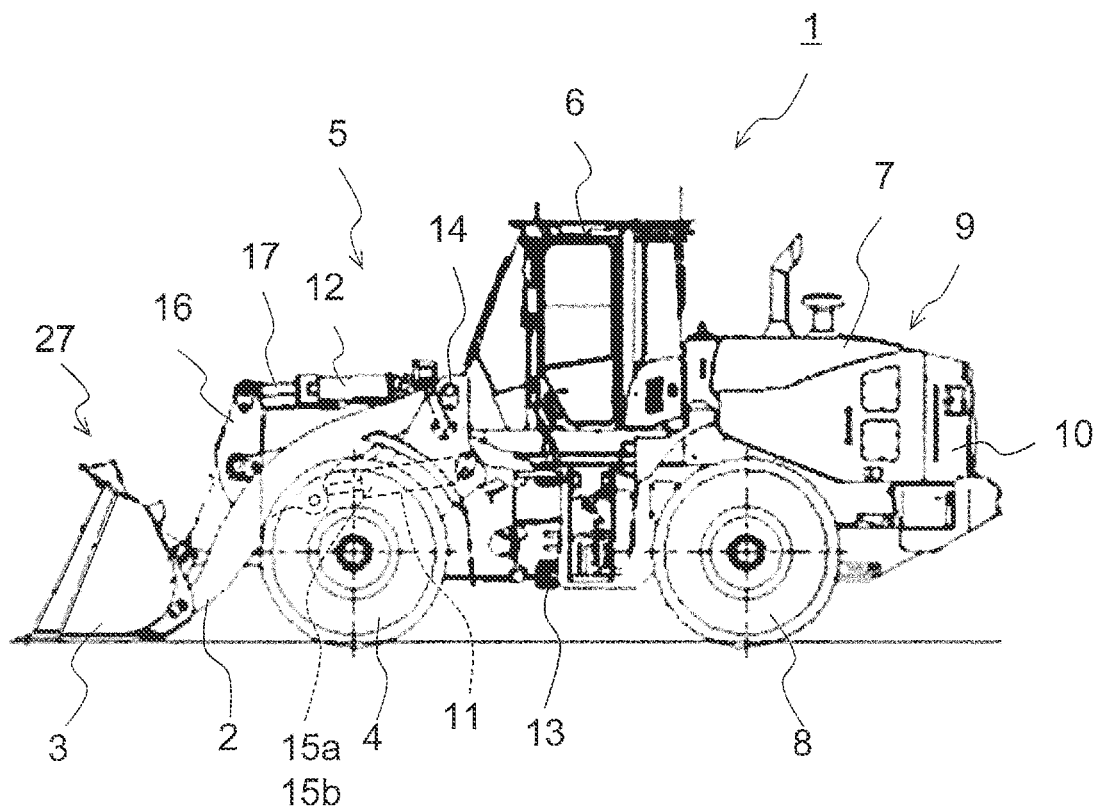


FIG. 2

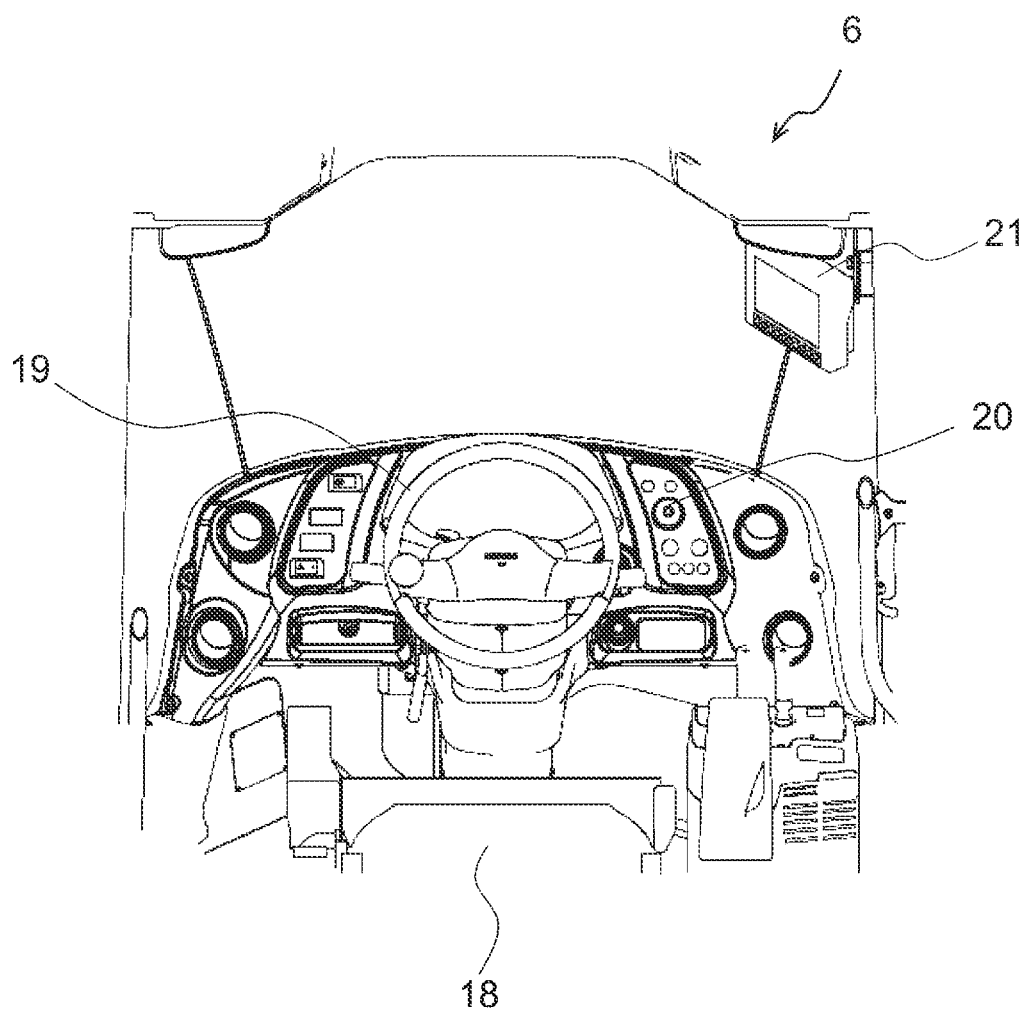


FIG. 3

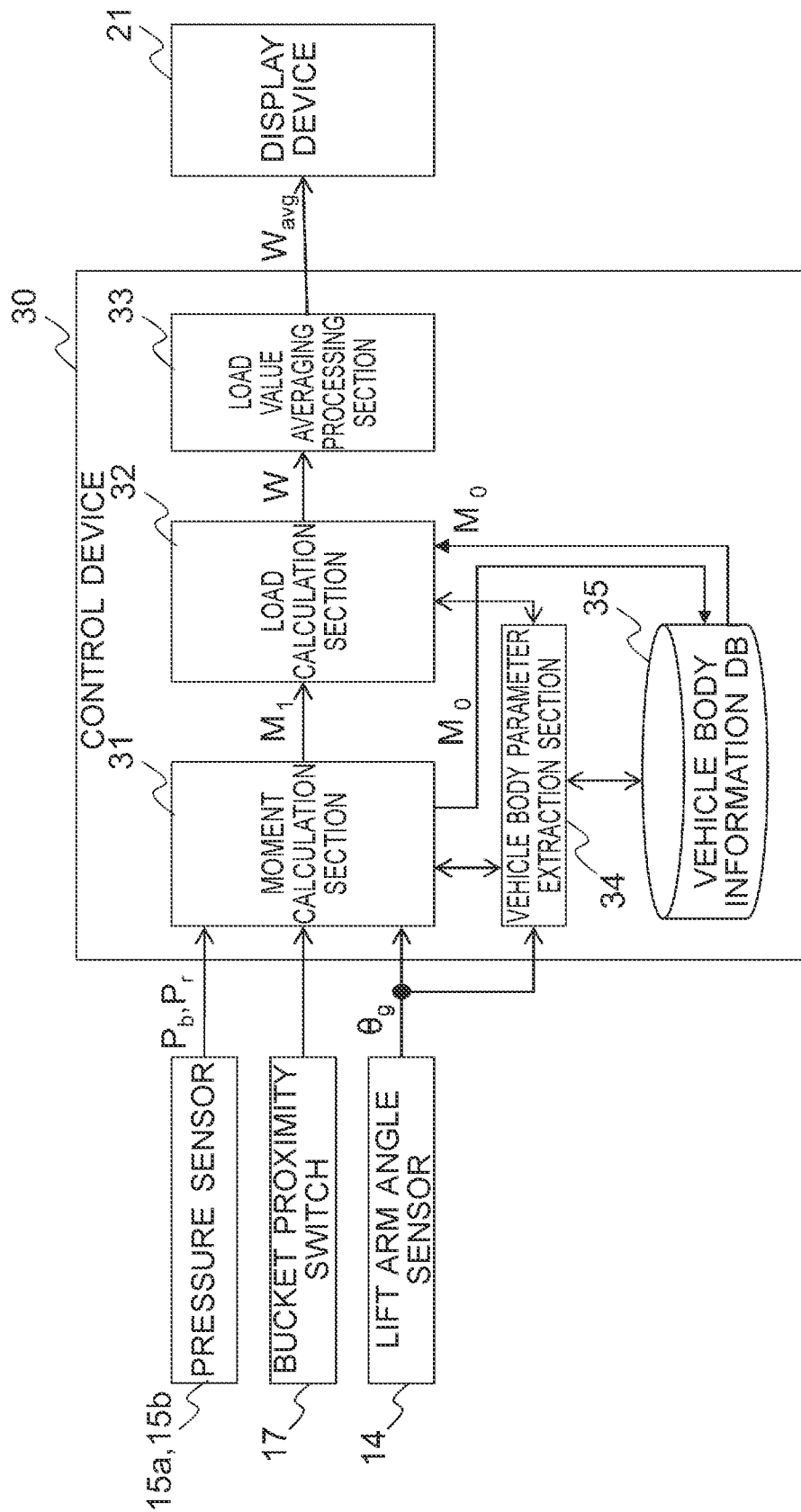


FIG. 5

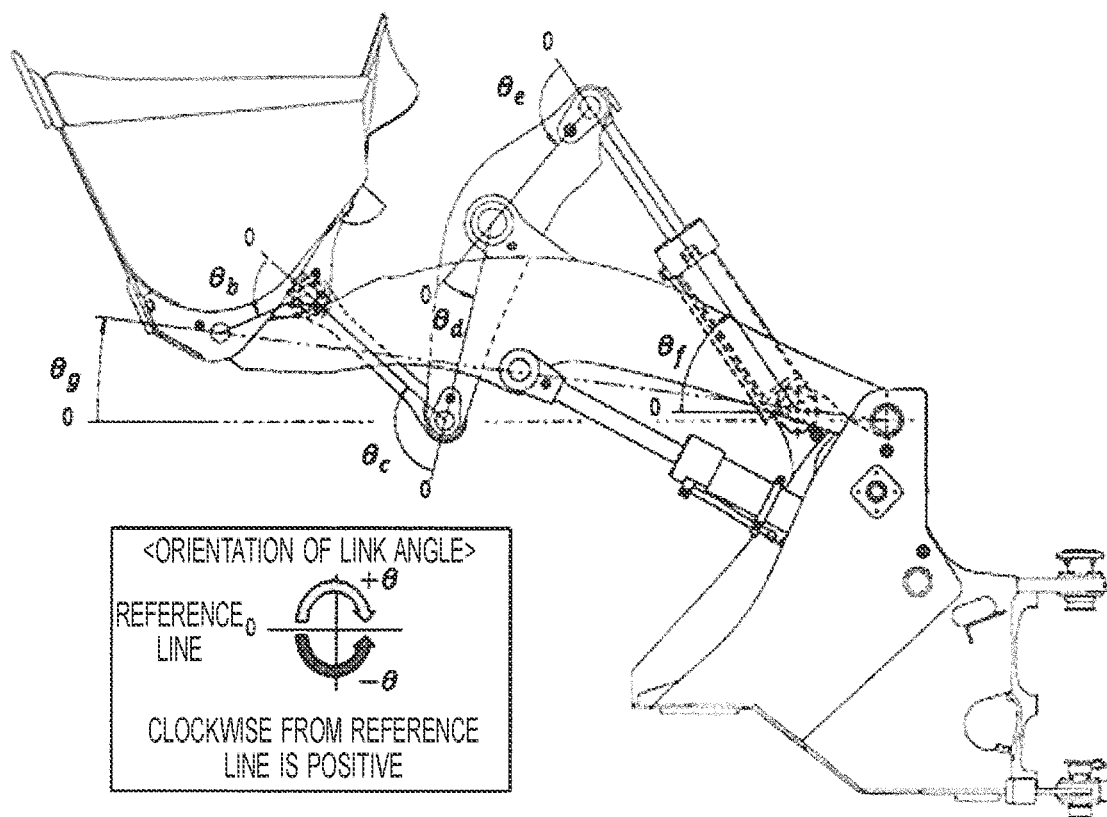
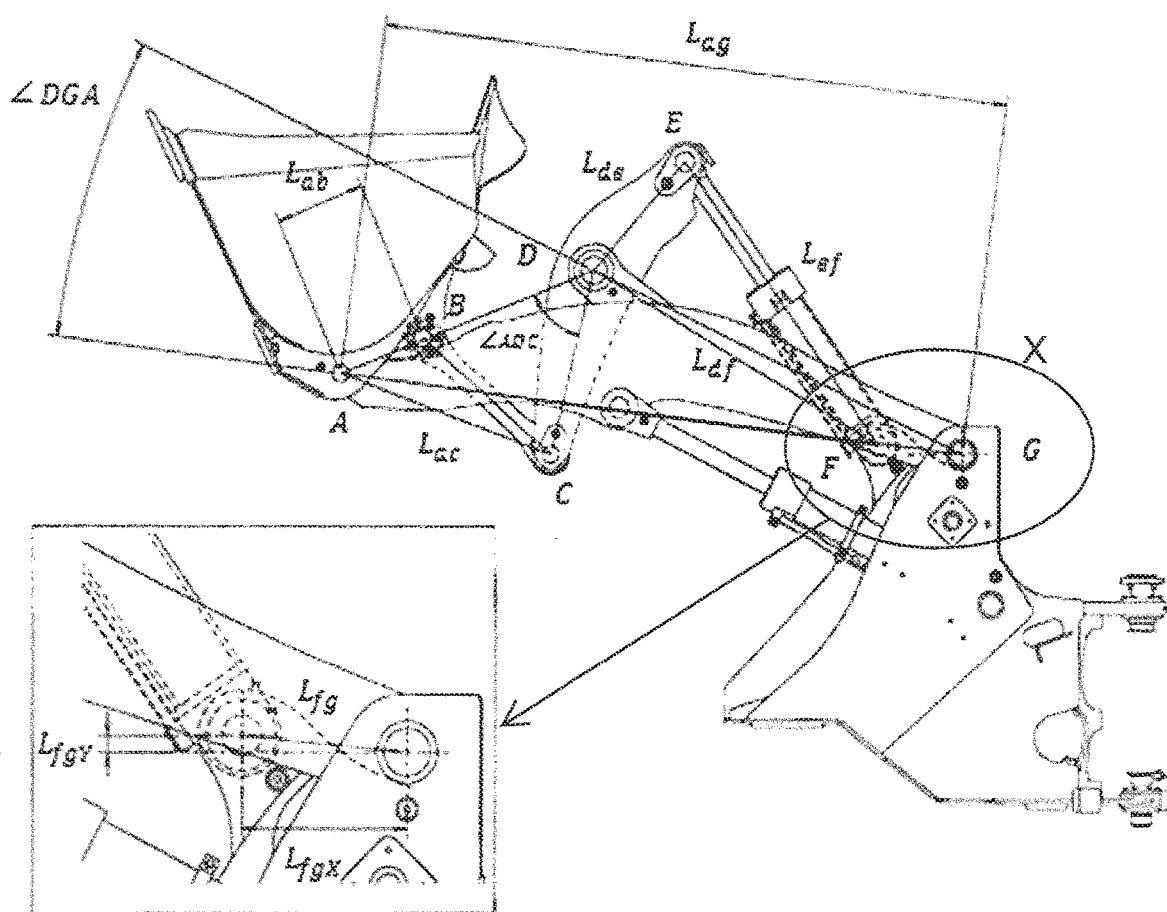


FIG. 6



DETAIL X

FIG. 7

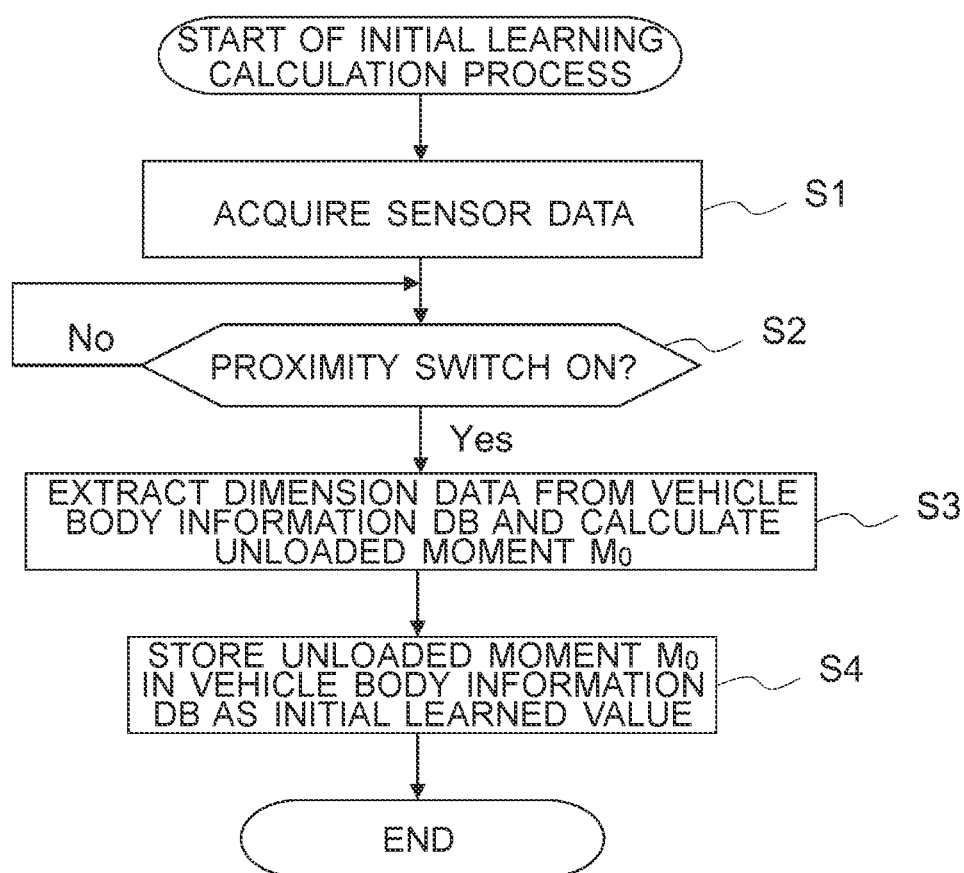


FIG. 8

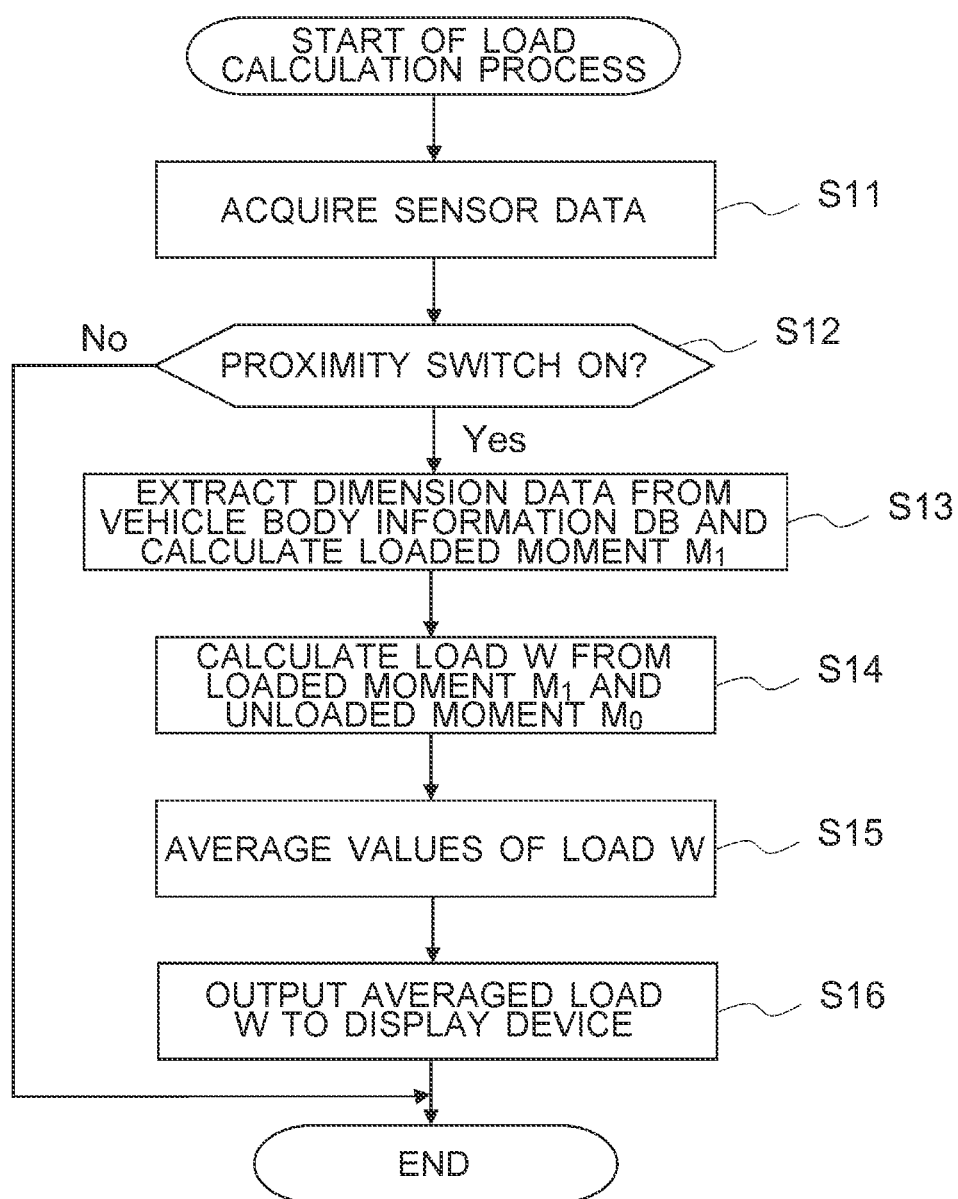
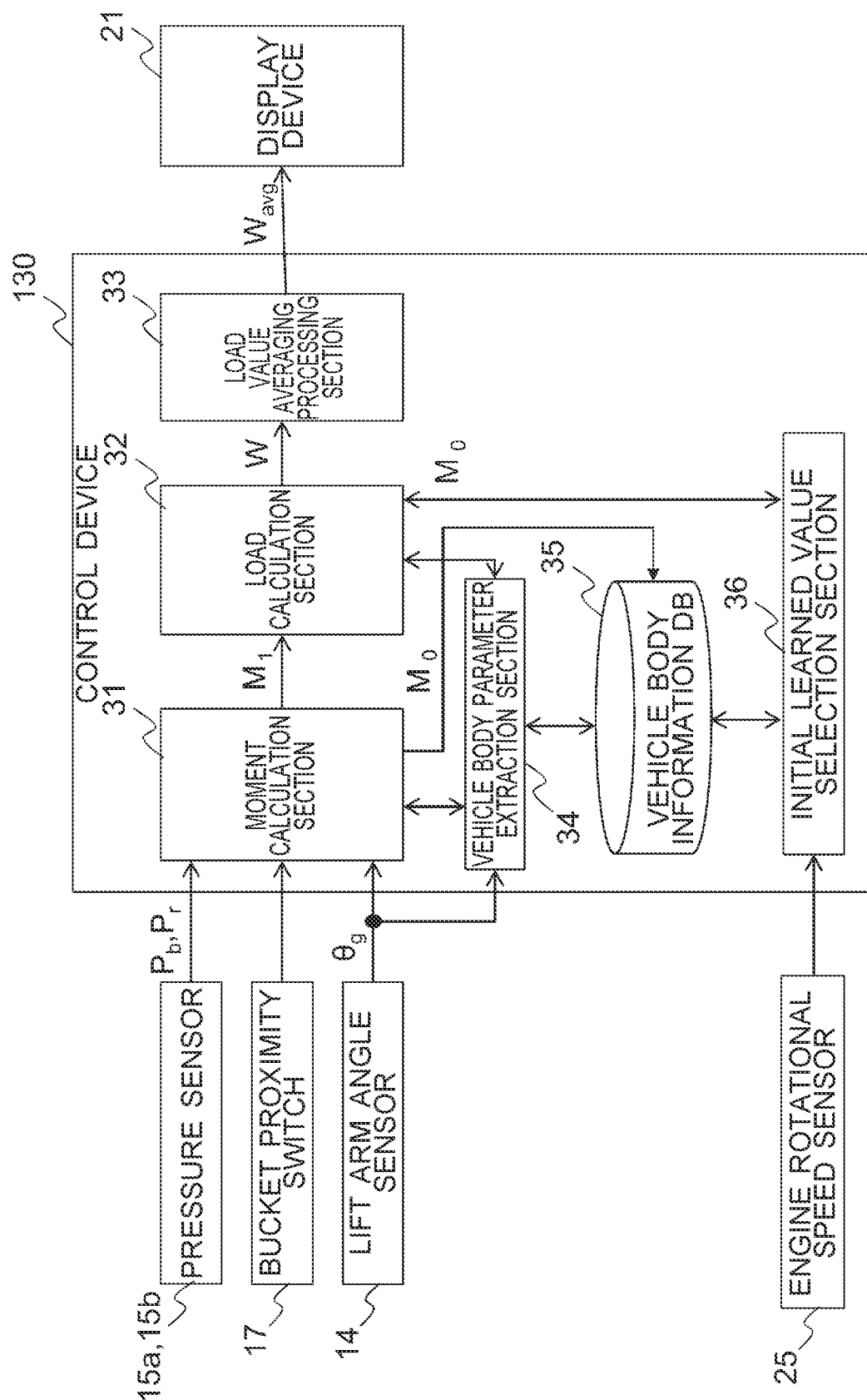


FIG. 9



WHEEL LOADER AND BUCKET LOAD CALCULATION METHOD

TECHNICAL FIELD

[0001] The present invention relates to a technical field for calculating a load applied to a bucket of a wheel loader.

BACKGROUND ART

[0002] A method described, for example, in Patent Literature 1 as a background art of the above-mentioned technical field “calculates the load of a material such as scrap metal or other scrap adhering to a front working device included in a magnet-type hydraulic excavator. More specifically, described is a load measurement method that calculates each moment in an unloaded state around first and second pins from the angle of each joint, the weight of each movable member, and gravity center position data on each movable member, calculates each moment in a loaded state around each pin from thrust calculated from a cylinder pressure around each pin, a boom angle, and a bucket angle, calculates a moment caused by a load around the first pin from the difference between the moment in a loaded state around the first pin and the moment in an unloaded state around the first pin, calculates a moment caused by a load around the second pin from the difference between the moment in a loaded state around the second pin and the moment in an unloaded state around the second pin, and calculates a lifting load by dividing the difference between the calculated moments by a horizontal distance between the first pin and the second pin.” (refer to Abstract).

CITATION LIST

Patent Literature

[0003] PATENT LITERATURE 1: Japanese Patent Application Laid-Open Publication No. 2006-78348

SUMMARY OF INVENTION

Technical Problem

[0004] A working device of a hydraulic excavator includes a boom, a boom cylinder, an arm, an arm cylinder, a bucket, and a bucket cylinder. The front working device described in Patent Literature 1 is configured such that a magnet is attached instead of a bucket. Meanwhile, a working device of a wheel loader includes a lift arm, a lift arm cylinder, a bucket, and a bucket cylinder. Thus, the elements of the working device of the wheel loader differ from those of the working device of the hydraulic excavator. Further, a link mechanism for rotating the bucket of the hydraulic excavator is different from that of the wheel loader.

[0005] The hydraulic excavator finally uses the boom cylinder to receive a load applied to the bucket. Therefore, the load applied to the bucket can be calculated from a cylinder pressure of the boom cylinder. However, the wheel loader uses the lift arm cylinder and the bucket cylinder to receive the load applied to the bucket. Therefore, the load measurement method for the hydraulic excavator cannot directly be applied to the wheel loader. Patent Literature 1 discloses the load measurement method for the hydraulic excavator, but describes nothing about the application of the load measurement method to the wheel loader.

[0006] An object of the present invention is to provide a wheel loader capable of accurately calculating a bucket load from the pressure of a lift arm cylinder and provide a bucket load calculation method suitable for the wheel loader.

Solution to Problem

[0007] In accomplishing the above object, according to a representative aspect of the present invention, there is provided a wheel loader including a vehicle body, a working device, a lift arm angle sensor, a pressure sensor, a control device, and a vehicle body information database. The working device includes a lift arm, a lift arm cylinder, a bucket, and a bucket cylinder. The lift arm is disposed at the front of the vehicle body and coupled to the vehicle body via a hinge pin. The lift arm cylinder rotates the lift arm in a vertical direction. The bucket is coupled to a tip of the lift arm. The bucket cylinder rotates the bucket in the vertical direction. The lift arm angle sensor detects the angle of the lift arm. The pressure sensor detects the pressure of the lift arm cylinder. The control device calculates a load on the bucket. The vehicle body information database stores vehicle information including dimension data on the working device. In an unloaded state of the bucket, the control device calculates an unloaded moment, namely, the moment around the hinge pin, from the pressure of the lift arm cylinder that is detected by the pressure sensor, the angle of the lift arm that is detected by the lift arm angle sensor, and the dimension data extracted from the vehicle body information database. In a loaded state of the bucket, the control device calculates a loaded moment, namely, the moment around the hinge pin, from the pressure of the lift arm cylinder that is detected by the pressure sensor, the angle of the lift arm that is detected by the lift arm angle sensor, and the dimension data extracted from the vehicle body information database. The control device calculates the load on the bucket by dividing the difference between the unloaded moment and the loaded moment by the horizontal distance between the hinge pin and the gravity center position of the bucket.

Advantageous Effects of Invention

[0008] The present invention makes it possible to accurately calculate a load on a bucket from the pressure of a lift arm cylinder. Problems, configurations, and advantageous effects other than those described above will become apparent from the following description of embodiments.

BRIEF DESCRIPTION OF DRAWINGS

[0009] FIG. 1 is a side view illustrating a wheel loader according to a first embodiment of the present invention.

[0010] FIG. 2 is a plan view illustrating the internal structure of a cab of the wheel loader depicted in FIG. 1.

[0011] FIG. 3 is a block diagram illustrating a control device according to the first embodiment.

[0012] FIG. 4 is a diagram illustrating a calculation model for calculating a load on a bucket.

[0013] FIG. 5 is a diagram illustrating a calculation model for calculating the load on the bucket.

[0014] FIG. 6 is a diagram illustrating a calculation model for calculating the load on the bucket.

[0015] FIG. 7 is a flowchart illustrating the steps of an initial learning calculation process.

[0016] FIG. 8 is a flowchart illustrating the steps of a load calculation process.

[0017] FIG. 9 is a block diagram illustrating the control device according to a second embodiment of the present invention.

DESCRIPTION OF EMBODIMENTS

[0018] Embodiments of a wheel loader according to the present invention will now be described with reference to the accompanying drawings.

First Embodiment

[0019] FIG. 1 is a side view illustrating a wheel loader 1 according to a first embodiment of the present invention. As illustrated in FIG. 1, the wheel loader 1 includes a working device 27, a front frame (vehicle body) 5 including, for example, tires 4, a cab 6, an engine compartment 7, and a rear frame (vehicle body) 9 including, for example, tires 8. An engine (not depicted) is mounted in the engine compartment 7. A counterweight 10 is mounted on the rear of the rear frame 9.

[0020] The working device 27 includes a lift arm 2, a lift arm cylinder 11, a bucket 3, and a bucket cylinder 12. The lift arm 2 is coupled to the front of the front frame 5 via a lift arm hinge pin G (see FIG. 4). The lift arm cylinder 11 pivots the lift arm 2 in a vertical direction. The bucket 3 is coupled to the tip of the lift arm 2. The bucket cylinder 12 pivots the bucket 3 in the vertical direction. The lift arm 2 includes a pair of left and right plate materials extended in a front-rear direction. The lift arm cylinder 11 includes a pair of left and right hydraulic cylinders.

[0021] The lift arm cylinder 11 drives the lift arm 2 in such a manner as to pivot the lift arm 2 in the vertical direction (elevation motion). The bucket cylinder 12 drives the bucket 3 in such a manner as to pivot the bucket 3 in the vertical direction (crowding or dumping). The front frame 5 and the rear frame 9 are pivotally coupled to each other by a center pin 13. A steering cylinder (not depicted) expands and contracts so as to bend the front frame 5 leftward and rightward with respect to the rear frame 9.

[0022] A lift arm angle sensor 14 is mounted at a joint between the lift arm 2 and the front frame 5, and used to detect the pivot angle of the lift arm 2. The lift arm cylinder 11 includes pressure sensors 15a, 15b. The pressure sensor 15a detects the bottom pressure of the lift arm cylinder 11. The pressure sensor 15b detects the rod pressure of the lift arm cylinder 11. A link mechanism including a bell crank 16 and a push rod 23 is disposed between the bucket cylinder 12 and the bucket 3. The bucket cylinder 12 pivots the bucket 3 via the link mechanism. The bucket cylinder 12 includes a bucket proximity switch 17. When the rod of the bucket cylinder 12 is placed in the most contracted state, the bucket proximity switch 17 turns on.

[0023] Although details will be described later, the present embodiment is configured so as to calculate the load W on the bucket 3 in response to detection signals from the pressure sensors 15a, 15b, the bucket proximity switch 17, and the lift arm angle sensor 14.

[0024] FIG. 2 is a plan view illustrating the internal structure of the cab 6. As illustrated in FIG. 2, for example, an operator seat 18, a steering wheel 19, a key switch 20, and a display device (monitor) 21 are installed in the cab 6. The operator seat 18 is a seat on which an operator sits. The steering wheel 19 controls the steering angle of the wheel loader 1. The key switch 20 starts and stops the wheel loader

1. The display device 21 presents information to the operator. The display device 21 displays not only the engine rotational speed and vehicle speed of the wheel loader 1, but also the load W on the bucket 3 that is calculated by a later-described control device 30.

[0025] The control device 30, which calculates the load W on the bucket 3, will now be described. The control device 30 includes, for example, a controller installed in the cab 6. The control device 30 performs various processes for calculating the load W on the bucket 3 and displaying a load W_{avg} (average value) on the display device 21. FIG. 3 is a block diagram illustrating the control device 30. As illustrated in FIG. 3, the control device 30 includes a moment calculation section 31, a load calculation section 32, a load value averaging processing section 33, a vehicle body parameter extraction section 34, and a vehicle body information database (DB) 35. The various processes to be performed by the control device 30 are implemented when a CPU (not depicted) loads a program stored, for example, in a hard disk drive (HDD) or other storage device into a memory and executes the loaded program. The vehicle body information database 35 is disposed in the storage device.

[0026] Signals and data to be inputted to the moment calculation section 31 are the detection signal of the lift arm angle sensor 14, the detection signals of the pressure sensors 15a, 15b, on/off signals of the bucket proximity switch 17, and dimension data on the working device 27 that is extracted by the vehicle body parameter extraction section 34. When the wheel loader 1 performs a normal operation, the moment calculation section 31 calculates a moment M_l (hereinafter referred to as the loaded moment M_l) around the lift arm hinge pin G in a loaded state of the bucket 3 in accordance with a later-described calculation formula. Further, when the wheel loader 1 performs an initial operation, the moment calculation section 31 calculates a moment M_o (hereinafter referred to as the unloaded moment M_o) around the lift arm hinge pin G in an unloaded state of the bucket 3, and stores the calculated unloaded moment M_o in the vehicle body information database 35 as an initial learned value.

[0027] The load calculation section 32 receives an input of the loaded moment M_l calculated by the moment calculation section 31, an input of the dimension data on the working device 27 that is extracted by the vehicle body parameter extraction section 34, and an input of the initial learned value (unloaded moment M_o), and then calculates the load W on the bucket 3 in accordance with a later-described calculation formula.

[0028] The load value averaging processing section 33 receives an input of the load W on the bucket 3, which is calculated by the load calculation section 32, and performs a process of averaging the inputted values of the load W on the bucket 3. The load value averaging processing section 33 then outputs data on the averaged load W_{avg} on the bucket 3 to the display device 21.

[0029] The vehicle body parameter extraction section 34 receives an input of the detection signal from the lift arm angle sensor 14, extracts, from the vehicle body information database 35, the dimension data on the working device 27 that is based on a lift arm angle 89 (see FIG. 5), and outputs the extracted dimension data to the moment calculation section 31.

[0030] The vehicle body information database 35 stores, for example, various dimension data on the working device

27, which are necessary for calculating the load W on the bucket 3, and the value of the unloaded moment M_0 (initial learned value).

[0031] A method of calculating the load W on the bucket 3 will now be described. FIGS. 4 to 6 are diagrams illustrating calculation models for calculating the load W on the bucket 3.

(Overview and Derivation of Moment-Based Load Calculation Formula)

[0032] For the derivation of a calculation formula for calculating the load W on the bucket 3 used in the present embodiment, it is necessary to obtain a relational expression indicative of the relationship between lift arm cylinder pressure and load W that is based on a moment M around the lift arm hinge pin G. For such a purpose, the relationship between force and moment applied to each cylinder 11, 12 is first clarified. The moment M around the lift arm hinge pin G is supported by the bucket cylinder 12 and the lift arm cylinder 11 as depicted in FIG. 4, and the associated relational expression is given by Formula (1) below.

[Formula 1]

$$M = F_c \times f + F_b \times e \quad (1)$$

where:

[0033] M=moment around lift arm hinge pin [Nm]

[0034] F_c =force applied to lift arm cylinder [N]

[0035] F_b =force applied to bucket cylinder [N]

[0036] f=horizontal length between hinge pin and vector of F_c [m]

[0037] e=horizontal length between hinge pin and vector of F_b [m]

[0038] Next, the relationship among the load W on the bucket 3, the horizontal length L_w between the lift arm hinge pin G and the gravity center position of the load W, the unloaded moment M_0 , and the loaded moment M_1 is expressed by Formula (2) below.

[Formula 2]

$$W = \frac{M_1 - M_0}{L_w} \quad (2)$$

where:

[0039] M_1 =moment around lift arm hinge pin in loaded state [Nm]

[0040] M_0 =moment around lift arm hinge pin in unloaded state [Nm]

[0041] W=load [kg]

[0042] L_w =horizontal length from gravity center position of load

[0043] The load W can be calculated by calculating the right side of Formula (2). A method of calculating the horizontal length L_w from the gravity center position of the load W will be described later.

(Moment Calculation Formula Based on Cylinder Bottom Pressure)

[0044] Next, a formula for calculating the moment M around the lift arm hinge pin G is derived from the bottom pressure and rod pressure of the lift arm cylinder 11, which

are actually measured by the pressure sensors 15a, 15b. First of all, a force F applied to the lift arm cylinder 11 can be calculated from the bottom pressure and rod pressure of the lift arm cylinder 11 by using Formula (3) below.

[Formula 3]

$$F = n(A_b \times P_b - A_r \times P_r) \quad (3)$$

where:

[0045] n=number of cylinders

[0046] A_b =bottom side pressure receiving area of cylinder [cm^2]

[0047] P_b =bottom pressure of cylinder [MPa]

[0048] A_r =rod side pressure receiving area of cylinder [cm^2]

[0049] P_r =rod pressure of cylinder [MPa]

[0050] Further, a force F_b applied to the bucket cylinder 12 can be expressed by Formula (4) below in accordance with the formula of equilibrium of moment.

[Formula 4]

$$F_b = W * \left(\frac{a}{b}\right) * \left(\frac{c}{d}\right) \quad (4)$$

where:

[0051] a=horizontal length between load center and bucket pin hinge pin [m]

[0052] b=distance between bucket pins [m]

[0053] c=distance between push rod center length and bell crank center pin [m]

[0054] d=distance between bell crank pins [m]

[0055] The loaded moment M_1 can be calculated from Formulas (1), (3), and (4) by using Formula (5) below. As $W=0$ when the bucket 3 is in an unloaded state, the influence on the bucket cylinder 12 is not taken into consideration when the unloaded moment M_0 is calculated.

[Formula 5]

$$M_1 = 2f(A_b * P_b - A_r * P_r) + W \left(\frac{ace}{bd}\right) \quad (5)$$

(Load Calculation Formula)

[0056] When Formula (5) is substituted into Formula (2), the relational expression between the lift arm cylinder pressure and the load W can be derived as expressed in Formula (6) below. The load W on the bucket 3 can be determined by solving Formula (6).

[Formula 6]

$$W = \frac{2f(A_b * P_b - A_r * P_r) - M_0}{L_w - \left(\frac{ace}{bd}\right)} \quad (6)$$

[0057] Referring to Formula (6), the bottom pressure of the lift arm cylinder 11, which is detected by the pressure sensor 15a, is substituted into P_b , the rod pressure of the lift arm cylinder 11, which is detected by the pressure sensor

15b, is substituted into P_r , the values of inter-link distances, which are determined from the dimension data (constant data) pre-stored in the vehicle body information database **35**, are substituted into a, b, c, d, e, f, and L_w , and the value of unloaded moment stored in the vehicle body information database **35** as an initial learned value at the time of initial operation of the wheel loader **1** is substituted into M_0 . For reducing the capacity of the storage device, the present embodiment is configured so that the value of the denominator of Formula (6), which is a variable, is stored in the vehicle body information database **35** as table data pre-associated with the value of the lift arm angle sensor **14** (lift arm angle θ_g). Therefore, when sensor data are inputted from the pressure sensors **15a**, **15b** and the lift arm angle sensor **14**, the control device **30** is able to calculate the load W on the bucket **3** in accordance with Formula (6).

[0058] Here, the inter-link distances a, b, c, d, e, f, L_w are variables dependent on the lift arm angle θ_g , and can be calculated by using a trigonometric function. A method of calculating the inter-link distances is described below.

[0059] First of all, variables and constants used in the following formulas are defined in Table 1. The constants listed in Table 1 are pre-stored in the vehicle body information database **35**. Meanwhile, the variables listed in Table 1 vary with the lift arm angle θ_g , and can be determined by the following formulas.

TABLE 1

Variable	Constant	Unit	Description
θ_g		deg	Lift arm angle $\angle$ (horizontal, Lag)
Ldf		mm	Length between bell crank D pin and bucket cylinder root F pin
	Ldg	mm	Length between bell crank D pin and lift arm hinge G pin
	Lfg	mm	Length between bucket cylinder root F pin and lift arm hinge G pin
	LfgX	mm	Horizontal length between bucket cylinder root F pin and lift arm hinge G pin
	LfgY	mm	Vertical length between bucket cylinder root F pin and lift arm hinge G pin
	$\angle DGA$	deg	$\angle$ (Ldg, Lag)
Laf		Mm	Length between lift arm tip A pin and bucket cylinder root F pin
	Lag	mm	Length between lift arm tip A pin and hinge G pin
	$\angle FGO$	deg	$\angle$ (Lfg, Horizontal)
θ_f		deg	Bucket cylinder posture angle $\angle$ (horizontal, Lef)
	Lef	mm	Bucket cylinder stroke length
θ_e		deg	Bell crank posture angle $\angle$ (Lde, Lef)
	Lde	mm	Bell crank DE pin length
$\angle ADC$		deg	$\angle$ (Lad, Lcd)
	Lad	mm	Length between lift arm tip A pin and center D pin
	θ_d	deg	$\angle$ (Lde, Lcd)
Lac		mm	Length between lift arm tip A pin and center C pin
	Lcd	mm	Bell crank CD pin length
θ_c		deg	$\angle$ (Lcd, Lbc)
	Lbc	mm	Distance between push rod BC pins
	Lab	mm	Distance between bucket AB pins
θ_b		deg	$\angle$ (Lbc, Lab)

(Inter-Link Distance Calculation Formula)

[0060] The inter-link distances a, b, c, d, e, f, L_w , are derived from formulas (7) to (13) by using a trigonometric function.

[Formula 7]

$$a = L_{loadG} \cos(\theta_f + \theta_e + \theta_d + \theta_c + \theta_b + 180 - H_{loadG}) \quad (7)$$

$$b = -L_{ab} \sin \theta_b \quad (8)$$

$$c = L_{cd} \sin \theta_e \quad (9)$$

$$d = -L_{de} \sin \theta_e \quad (10)$$

$$e = L_{fg} \sin \{ (180 - \theta_g) + \angle FGO \} \quad (11)$$

$$f = L_{gi} \sin \theta_i \quad (12)$$

$$L_w = L_{ag} \cos \theta_g + L_{loadG} \cos(\theta_f + \theta_e + \theta_d + \theta_c + \theta_b + 180 - H_{loadG}) \quad (13)$$

where:

[0061] L_{loadG} =load center length

[0062] H_{loadG} =load center angle

[0063] H_{bmcyl} =lift arm cylinder angle

[0064] However, θ_i is determined by formula (14).

[Formula 8]

$$\theta_i = \tan^{-1} \left\{ \frac{L_{gi} \sin(\theta_g + H_{bmcyl}) + L_{ghY}}{L_{gi} \cos(\theta_g + H_{bmcyl}) + L_{ghX}} \right\} - (\theta_g + H_{bmcyl}) \quad (14)$$

(Link Posture Angle Calculation Formula)

[0065] A method of calculating individual link posture angles (hereinafter referred to as the link angles) θ_f , θ_e , θ_c ,

θ_b used in Formulas (7) to (13) above will now be described. Each of the link angles can be geometrically calculated by using the law of cosines in accordance with the dimension data (constant) on the working device **27**, which is stored in the vehicle body information database **35**, and with the lift arm angle **89** detected by the lift arm angle sensor **14**.

[0066] The inter-link distances L_{df} , L_{af} are determined by the law of cosines as expressed in Formulas (15) and (16), respectively.

[Formula 9]

$$L_{df} = \sqrt{L_{dg}^2 + L_{fg}^2 - 2L_{dg} \{L_{fg} \cos(\theta_g + \angle DGA) + L_{fg} \sin(\theta_g + \angle DGA)\}} \quad (15)$$

$$L_{ef} = \sqrt{L_{og}^2 + L_{fg}^2 - 2L_{og} L_{fg} \cos(\theta_g - \angle FGO)} \quad (16)$$

[0067] In this instance, the link angles θ_f , θ_e are derived as expressed in Formulas (17) and (18), respectively.

[Formula 10]

$$\theta_f = \tan^{-1} \left\{ \frac{L_{dg} \sin(\theta_g + \angle DGA) - L_{fg} \gamma}{L_{dg} \cos(\theta_g + \angle DGA) - L_{fg} \chi} \right\} + \cos^{-1} \left(\frac{L_{df}^2 + L_{ef}^2 - L_{de}^2}{2L_{df} L_{ef}} \right) \quad (17)$$

$$\theta_e = \cos^{-1} \left(\frac{L_{de}^2 + L_{ef}^2 - L_{df}^2}{2L_{de} L_{ef}} \right) - 180 \quad (18)$$

[0068] Further, based on the inter-link distances L_{df} , L_{af} calculated from Formulas (15) and (16), the angle ADC and the distance L_{ac} are determined by the law of cosines as expressed in Formulas (19) and (20), respectively.

[Formula 11]

$$\angle ADC = \cos^{-1} \left(\frac{L_{ad}^2 + L_{df}^2 - L_{af}^2}{2L_{ad} L_{df}} \right) + \cos^{-1} \left(\frac{L_{de}^2 + L_{df}^2 - L_{ef}^2}{2L_{de} L_{df}} \right) - 180 - \theta_d \quad (19)$$

$$L_{ac} = \sqrt{L_{ad}^2 + L_{cd}^2 - 2L_{ad} L_{cd} \cos \angle ADC} \quad (20)$$

[0069] In this instance, the link angles θ_c , θ_b are derived as expressed in Formulas (21) and (22), respectively.

[Formula 12]

$$\theta_c = 180 + \cos^{-1} \left(\frac{L_{ac}^2 + L_{bc}^2 - L_{ab}^2}{2L_{ac} L_{bc}} \right) - \cos^{-1} \left(\frac{L_{cd} - L_{ad} \cos \angle ADC}{L_{ac}} \right) \quad (21)$$

$$\theta_b = \cos^{-1} \left(\frac{L_{ab}^2 + L_{bc}^2 - L_{ac}^2}{2L_{ab} L_{bc}} \right) - 180 \quad (22)$$

[0070] When the variables determined by Formulas (15) to (22) are substituted into Formulas (7) to (13), the inter-link distances a , b , c , d , e , f , L_w are obtained.

[0071] Steps of calculation processes performed by the control device 30 will now be described. First of all, an initial learning calculation process to be performed at the time of initial operation of the wheel loader 1 will be described. FIG. 7 is a flowchart illustrating the steps of the

initial learning calculation process. The process depicted in FIG. 7 is performed only in a case where, for example, the control device 30 is placed in an initial learning setting mode when the wheel loader 1 is to be shipped. The process starts when the key switch 20, which starts and stops the wheel loader 1, is turned on. Switching into the initial learning setting mode can be made when, for example, the operator outputs a mode change command from a dedicated terminal device to the control device 30.

[0072] As illustrated in FIG. 7, the moment calculation section 31 acquires sensor data from the pressure sensors 15a, 15b and the lift arm angle sensor 14 (step S1). Upon receipt of an ON signal inputted from the bucket proximity switch 17 (“YES” at step S2), the moment calculation section 31 determines that the bucket 3 is in a posture ready to be loaded with earth, sand, and other materials, and requests the vehicle body parameter extraction section 34 to extract dimension data on the working device 27. Based on the lift arm angle θ_g received from the lift arm angle sensor 14, the vehicle body parameter extraction section 34 extracts the dimension data corresponding to the lift arm angle θ_g from the vehicle body information database 35, and outputs the extracted dimension data to the moment calculation section 31. Based on the inputted sensor data and dimension data, the moment calculation section 31 calculates the unloaded moment M_o (step S3). The moment calculation section 31 then stores the unloaded moment M_o in the vehicle body information database 35 as the initial learned value (step S4). If the ON signal is not inputted from the bucket proximity switch 17 (“NO” at step S2), the moment calculation section 31 stands by in step S2 until the ON signal is inputted.

[0073] A load calculation process of calculating the load W on the bucket 3 will now be described. FIG. 8 is a flowchart illustrating the steps of the load calculation process. The process depicted in FIG. 8 starts when the key switch 20, which starts and stops the wheel loader 1, is turned on at the time of normal operation. The process is repeatedly executed at predetermined intervals (e.g., at intervals of several seconds) by performing steps S1 to S4.

[0074] As illustrated in FIG. 8, the moment calculation section 31 acquires sensor data from the pressure sensors 15a, 15b and the lift arm angle sensor 14 (step S11). Upon receipt of an ON signal inputted from the bucket proximity switch 17 (“YES” at step S12), the moment calculation section 31 determines that the bucket 3 is in a posture ready to be loaded with earth, sand, and other materials, and requests the vehicle body parameter extraction section 34 to extract dimension data on the working device 27. Based on the lift arm angle θ_g received from the lift arm angle sensor 14, the vehicle body parameter extraction section 34 extracts the dimension data corresponding to the lift arm angle θ_g from the vehicle body information database 35, and outputs the extracted dimension data to the moment calculation section 31. Based on the inputted sensor data and dimension data, the moment calculation section 31 calculates the loaded moment M_l , and outputs the calculated loaded moment M_l to the load calculation section 32 (step S13).

[0075] The load calculation section 32 acquires the dimension data stored in the vehicle body information database 35 via the vehicle body parameter extraction section 34, and additionally acquires the unloaded moment M from the vehicle body information database 35. Then, by using Formula (6) above, the load calculation section 32 calculates the

load W on the bucket **3** from the difference between the loaded moment M_l and the unloaded moment M_o , and outputs the calculated load W to the load value averaging processing section **33** (step **S14**).

[0076] The load value averaging processing section **33** averages the values of the load W periodically inputted from the load calculation section **32** (step **S15**), and outputs the averaged load W_{avg} to the display device **21** (step **S16**). Accordingly, the display device **21** displays the load on the bucket **3** during a loading operation of the wheel loader **1**. If the ON signal is not inputted from the bucket proximity switch **17** ("NO" at step **S12**), the process terminates.

[0077] As described above, the first embodiment uses the above-described calculation formulas for the wheel loader **1**, which supports the load on the bucket **3** with the lift arm cylinder **11** and the bucket cylinder **12**. Therefore, the first embodiment makes it possible to accurately calculate the load W on the bucket **3** from the dimension data on the working device **27**, the bottom pressure and rod pressure of the lift arm cylinder **11**, and the angle of the lift arm **2** without having to detect the pressure of the bucket cylinder **12**.

[0078] Further, conventional wheel loaders, in most cases, do not include a sensor for detecting the pressure of the bucket cylinder **12**. Meanwhile, the present embodiment eliminates the necessity of detecting the pressure of the bucket cylinder **12**. Therefore, the load W on the bucket **3** can easily be calculated simply by incorporating a load calculation program according to the present invention into a controller for a conventional wheel loader.

[0079] Furthermore, when the unloaded moment M_o is determined once at the time of initial operation of the wheel loader **1**, the present embodiment makes it possible to calculate the load W on the bucket **3** and display the calculated load W on the display device **21**. In the past, it was necessary to load a bucket with a calibration weight or other load whose weight is accurately known, and perform initial learning of the relationship among lift arm cylinder pressure, lift arm angle, and load weight. Further, it was necessary to perform such initial learning a number of times while varying the weight of the load. Thus, the initial learning was extremely troublesome. Additionally, in a case where a bucket was field-replaced, it was necessary to perform the similar initial learning. In most cases, however, a calibration weight or other heavy load could not be prepared in the field. This made it difficult to perform the initial learning.

[0080] Meanwhile, the wheel loader **1** according to the first embodiment is configured so that the control device **30** automatically calculates the unloaded moment M_o when the lift arm **2** is raised to keep the bucket **3** in a crowding posture. This eliminates the necessity of performing initial learning by using a calibration weight or other similar load. Consequently, even in a case where the bucket **3** is field-replaced, the load W on the newly employed bucket **3** can be accurately calculated as far as the unloaded moment M_o is calculated once, at the time of an initial operation subsequent to bucket replacement, from the bottom pressure and rod pressure of the lift arm cylinder **11**, which are detected by the pressure sensors **15a**, **15b**, and the lift arm angle **89** detected by the lift arm angle sensor **14**.

Second Embodiment

[0081] The wheel loader according to a second embodiment of the present invention will now be described. The second embodiment differs from the first embodiment in the configuration of the control device. Therefore, the following description mainly deals with the difference in the control device, and elements identical with those in the first embodiment will not be redundantly described.

[0082] FIG. **9** is a block diagram illustrating the control device **130** according to the second embodiment. As illustrated in FIG. **9**, the wheel loader according to the second embodiment includes an engine rotational speed sensor **25** in order to detect the rotational speed of the engine. Thus, the second embodiment differs from the first embodiment in that the vehicle body information database **35** stores a plurality of values of the unloaded moment M_o corresponding to various engine rotational speeds. Therefore, as the initial learning, the moment calculation section **31** calculates the plurality of values of the unloaded moment M_o for various engine rotational speeds, and stores the calculated values of the unloaded moment M_o in the vehicle body information database **35**. When calculating the load W on the bucket **3**, the load calculation section **32** requests an initial learned value selection section **36** to extract the unloaded moment M_o corresponding to an engine rotational speed inputted from the engine rotational speed sensor **24**. The load calculation section **32** then calculates the load W by using the unloaded moment M_o corresponding to the engine rotational speed outputted from the initial learned value selection section **36**.

[0083] The second embodiment permits the use of the unloaded moment M_o corresponding to the engine rotational speed and thus makes it possible to calculate the load W on the bucket **3** more accurately than the first embodiment. More specifically, the bottom pressure and rod pressure of the lift arm cylinder **11**, which are detected by the pressure sensors **15a**, **15b**, also vary with the engine rotational speed. Consequently, when a plurality of values of the unloaded moment M_o corresponding to various values of the engine rotational speed are stored in the vehicle body information database **35**, the load W on the bucket **3** can be calculated more accurately in consideration of the engine rotational speed.

[0084] The foregoing embodiments are merely illustrative for explaining the present invention and are not intended to limit the scope of the present invention to the foregoing embodiments alone. Persons skilled in the art can implement the present invention in various other modes without departing from the spirit of the present invention.

LIST OF REFERENCE SIGNS

- [0085]** 1: Wheel loader
- [0086]** 2: Lift arm
- [0087]** 3: Bucket
- [0088]** 5: Front frame (vehicle body)
- [0089]** 9: Rear frame (vehicle body)
- [0090]** 11: Lift arm cylinder
- [0091]** 12: Bucket cylinder
- [0092]** 14: Lift arm angle sensor
- [0093]** 15a, 15b: Pressure sensor
- [0094]** 16: Bell crank
- [0095]** 17: Bucket proximity switch
- [0096]** 21: Display device

[0097] 23: Push rod
 [0098] 25: Engine rotational speed sensor
 [0099] 30: Control device
 [0100] 31: Moment calculation section
 [0101] 32: Load calculation section
 [0102] 33: Load value averaging processing section
 [0103] 34: Vehicle body parameter extraction section
 [0104] 35: Vehicle body information database (DB)
 [0105] 36: Initial learned value selection section
 [0106] 130: Control device

1. A wheel loader comprising:

a vehicle body;

a working device that includes a lift arm, a lift arm cylinder, a bucket, and a bucket cylinder, the lift arm being disposed at the front of the vehicle body and coupled to the vehicle body via a hinge pin, the lift arm cylinder rotating the lift arm in a vertical direction, the bucket being coupled to a tip of the lift arm, the bucket cylinder rotating the bucket in the vertical direction;

a lift arm angle sensor that detects the angle of the lift arm;

a pressure sensor that detects the pressure of the lift arm cylinder;

a control device that calculates a load on the bucket; and
 a vehicle body information database that stores vehicle body information including dimension data on the working device,

wherein, in an unloaded state of the bucket, the control device calculates an unloaded moment, namely, the moment around the hinge pin, from the pressure of the lift arm cylinder that is detected by the pressure sensor, the angle of the lift arm that is detected by the lift arm angle sensor, and the dimension data extracted from the vehicle body information database,

in a loaded state of the bucket, the control device calculates a loaded moment, namely, the moment around the hinge pin, from the pressure of the lift arm cylinder that is detected by the pressure sensor, the angle of the lift arm that is detected by the lift arm angle sensor, and the dimension data extracted from the vehicle body information database, and

the control device calculates the load on the bucket by dividing the difference between the unloaded moment and the loaded moment by the horizontal distance between the hinge pin and the gravity center position of the bucket.

2. The wheel loader according to claim 1,

wherein, at the time of initial operation of the wheel loader, the control device calculates the unloaded moment and stores the calculated unloaded moment in the vehicle body information database, and

when calculating the load on the bucket at the time of normal operation of the wheel loader, the control device calculates the load on the bucket from the unloaded moment and the loaded moment, the unloaded moment being stored in the vehicle body information database, the loaded moment being calculated based on measured data from the pressure sensor and the lift arm angle sensor.

3. The wheel loader according to claim 1, further comprising:

a bucket proximity switch that turns on when a rod of the bucket cylinder is in the most contracted state,

wherein the control device calculates the unloaded moment and the loaded moment in response to an ON signal from the bucket proximity switch.

4. The wheel loader according to claim 2, further comprising:

an engine that drives the vehicle body to travel; and

an engine rotational speed sensor that detects the rotational speed of the engine,

wherein, at the time of initial operation of the wheel loader, the control device calculates a plurality of values of the unloaded moment corresponding to a plurality of values of the engine rotational speed and stores the calculated values in the vehicle body information database in association with the values of the engine rotational speed, and

when calculating the load on the bucket at the time of normal operation of the wheel loader, the control device calculates the load on the bucket from the unloaded moment and the loaded moment, the unloaded moment being corresponding to the engine rotational speed detected by the engine rotational speed sensor, the loaded moment being calculated based on a measured value.

5. The wheel loader according to claim 2, further comprising:

a bucket proximity switch that turns on when a rod of the bucket cylinder is in the most contracted state,

wherein the control device calculates the unloaded moment and the loaded moment in response to an ON signal from the bucket proximity switch.

6. A bucket load calculation method that is applied to a wheel loader and used to calculate a load on a bucket, the wheel loader including a vehicle body, a working device, a lift arm angle sensor, a pressure sensor, and a vehicle body information database, the working device including a lift arm, a lift arm cylinder, a bucket, and a bucket cylinder, the lift arm being disposed at the front of the vehicle body and coupled to the vehicle body via a hinge pin, the lift arm cylinder rotating the lift arm in a vertical direction, the bucket being coupled to a tip of the lift arm, the bucket cylinder rotating the bucket in the vertical direction, the lift arm angle sensor detecting the angle of the lift arm, the pressure sensor detecting the pressure of the lift arm cylinder, the vehicle body information database storing various vehicle body information including dimension data on the working device, the bucket load calculation method comprising:

a first step of calculating, in an unloaded state of the bucket, an unloaded moment, namely, the moment around the hinge pin, from the pressure of the lift arm cylinder that is detected by the pressure sensor, the angle of the lift arm that is detected by the lift arm angle sensor, and the dimension data extracted from the vehicle body information database, and storing the calculated unloaded moment in the vehicle body information database as an initial learned value;

a second step of calculating, in a loaded state of the bucket, a loaded moment, namely, the moment around the hinge pin, from the pressure of the lift arm cylinder that is detected by the pressure sensor, the angle of the lift arm that is detected by the lift arm angle sensor, and the dimension data extracted from the vehicle body information database; and

a third step of calculating the load on the bucket by dividing the difference between the unloaded moment and the loaded moment by the horizontal distance between the hinge pin and the gravity center position of the bucket, the unloaded moment being stored in the vehicle body information database in the first step, the loaded moment being calculated in the second step.

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